

perdas provenientes da emissão espontânea dos átomos no estado excitado...

Problema principal

O que acontece quando olhar num feixe laser?

1mW

Laser

He - Ne



(continuação depois)

1ª pergunta: O que acontece quando olhamos diretamente para o Sol?

(D_{Sol})
diâmetro



$1.4 \times 10^9 \text{ m}$

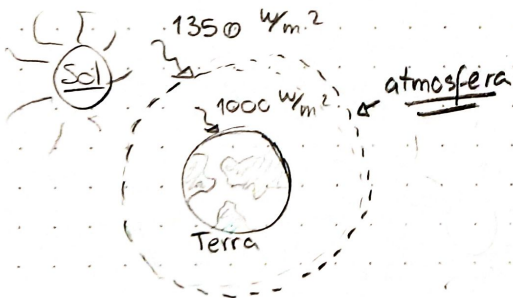
$I_{\text{Retina}} = ?$

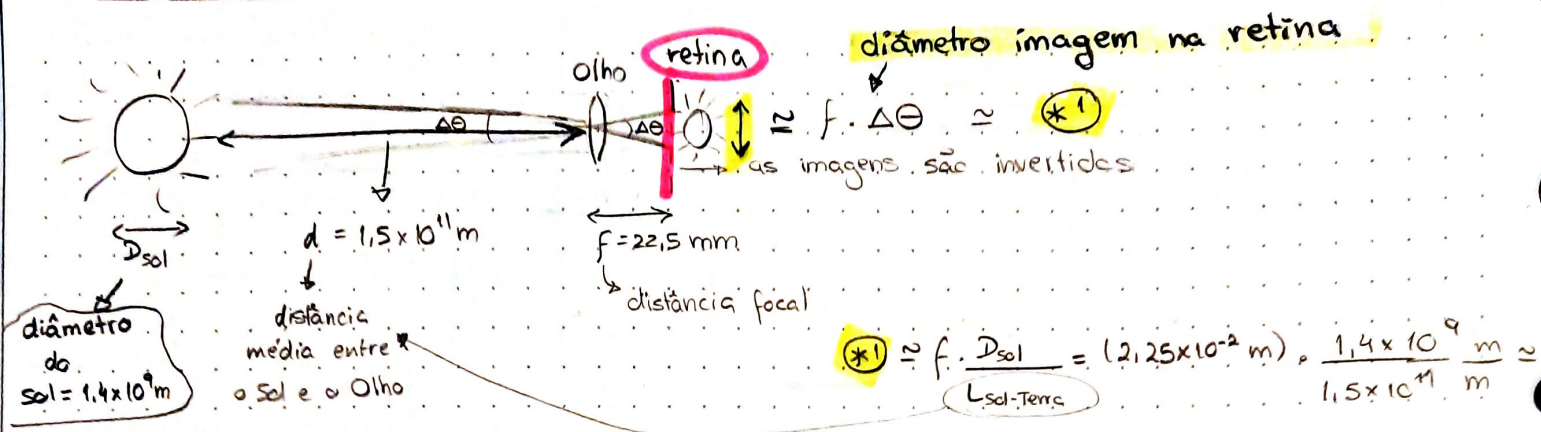
↓
Irradiância



pupila sobre
iluminação forte:
diâmetro $\approx 2 \text{ mm}$

$$P_{\text{sol}} \rightarrow \text{olho} = 1000 \frac{\text{W}}{\text{m}^2} \times (\pi (10^{-3} \text{ m})^2) =$$
$$= \pi \times 10^{-3} \text{ W} = \underline{\underline{\pi \text{ mW}}}$$





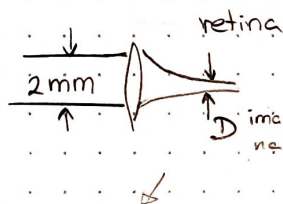
$$I_{\text{Sol}}^{\text{Retina}} = \frac{P_{\text{Sol}}}{\pi \left(\frac{D_{\text{imagem do Sol na retina}}}{2} \right)^2} \approx 10^5 \frac{\text{W}}{\text{m}^2}$$

$\phi, 2 \text{ mm}$

$D_{\text{imagem do Sol na retina}}$

Voltando ao Problema Principal:

$$P_{\text{laser}}^{\text{olho}} = 1 \text{ mW}$$



depende do comp. focal

$$I_{\text{laser}}^{\text{retina}} = \frac{P_{\text{laser}}}{\pi \left(\frac{D_{\text{imagem do laser na retina}}}{2} \right)^2} = \frac{1 \text{ mW}}{\pi (4,5 \times 10^{-6} \text{ m})^2} = 1,5 \times 10^7 \text{ W/m}^2$$

150 x maior que $I_{\text{Sol}}^{\text{Retina}}$

Recorda:

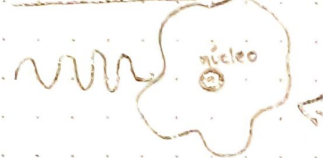
$$P_{\text{spot}} = 122 \cdot f \cdot \frac{\lambda}{D}$$

$$= \frac{\lambda}{d_{\text{laser}}} \cdot \frac{4}{\pi} \cdot f = \frac{632,8 \text{ nm}}{2 \text{ mm}} \cdot \frac{4}{\pi} \cdot 22,5 \text{ mm} = 9 \mu\text{m}$$

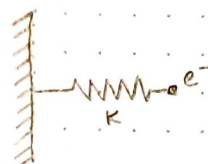
Resolução de exercícios:

Problema 3.3. ^{do livro}

Modelo de Lorentz



Lorentz



frequência atômica

$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \cdot \sqrt{\frac{K}{m}}$$

$$\nu_{\text{átomo}} \sim 5 \times 10^{14} \text{ Hz}$$

$$\lambda = \frac{c}{\nu} = 600 \text{ nm}$$

corresponde a 374 nm

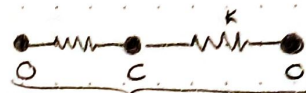
corresponde a 300 nm

$$\nu_{\text{moléculas}} \sim 8 \times 10^{11} - 10^{13} \text{ Hz}$$

$$\frac{\nu_{\text{molecular}}}{\nu_{\text{atômica}}} = \sqrt{\frac{m_{e^-}}{m_{\text{átomo}}}}$$

(*) → segundo o modelo de Lorentz, podemos considerar a molécula assim.

Se estivermos a falar de uma molécula:



(*)

frequência molecular

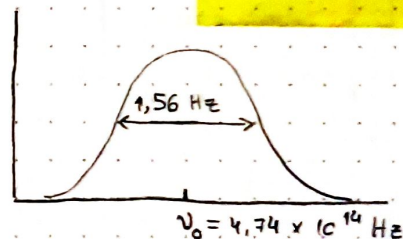
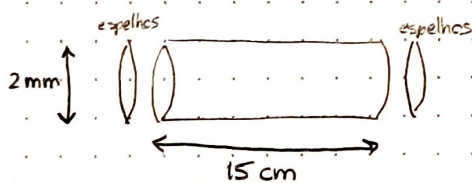
$$\nu_{\text{molecular}} = \frac{1}{2\pi} \cdot \sqrt{\frac{K}{m_{\text{molécula}}}}$$

$$m_{\text{átomos}} \sim (1 - 200) \cdot m_{\text{protão}}$$

$$1,67 \times 10^{-27} \text{ kg}$$

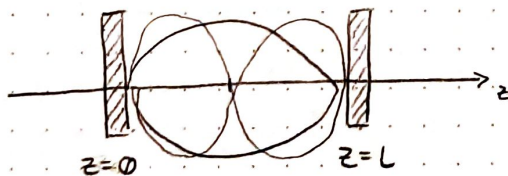
$$m_{e^-} = 9,1 \times 10^{-31} \text{ kg}$$

Problema stor 1: Laser He-Ne, imaginando um tubo com espelhos nas faces laterais, o perfil transição laser está centrado da forma:



Caso 1: cavidade 1ª dimensão

$$\lambda = 632,8 \text{ nm}$$



$$n \cdot \frac{\lambda_n}{2} = L$$

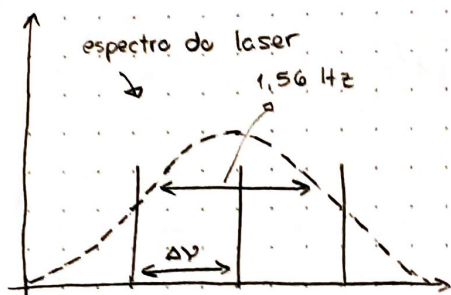
$$\pi \cdot n = \frac{2\pi \cdot L}{\lambda_n} = k_n \cdot L$$

$$c = \frac{\omega}{k} = \frac{2\pi \cdot \nu}{k} \rightarrow k = \frac{2\pi \cdot \nu}{c}$$

$$\pi \cdot n = \frac{2\pi \cdot \nu_n \cdot L}{c}$$

$$\nu_n = n \cdot \left[\frac{c}{2L} \right]$$

$$\Delta \nu = \frac{c}{2L} = \frac{3 \times 10^8 \text{ m/s}}{0,3 \text{ m}} = 10^9 \text{ Hz}$$

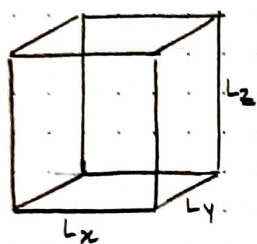


$$\frac{1,56 \text{ Hz}}{16,4 \text{ Hz}} = 1,5 \text{ nodos}$$

ou seja 1 ou 2

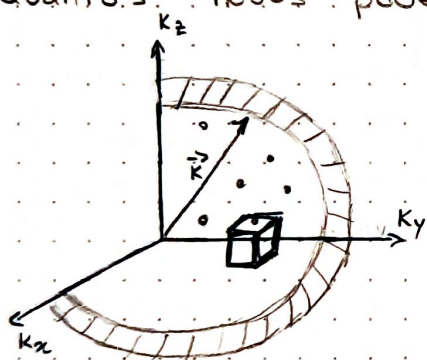
isto vem de: $n^\circ \text{ nodos} = \frac{\Delta \nu_{\text{laser}}}{\Delta \nu_{\text{cavidade}}}$

Se falarmos agora em 3 dimensões:



$$\left. \begin{aligned} k_x \cdot L_x &= n_x \cdot \pi \\ k_y \cdot L_y &= n_y \cdot \pi \\ k_z \cdot L_z &= n_z \cdot \pi \end{aligned} \right\} \begin{array}{l} \text{e agora faz-se o} \\ \text{que se fez em 1D!} \end{array}$$

Quanto nodos podem existir entre $|\vec{k}|$ e $|\vec{k}| + |d\vec{k}|$?
(para $k > 0$:)



$$n^\circ \text{ nodos} = \frac{2 \cdot \left(\frac{1}{8} \cdot 4 \cdot \pi \cdot k^2 \cdot dk \right)}{\underbrace{\left(\frac{\pi}{L_x} \right) \cdot \left(\frac{\pi}{L_y} \right) \cdot \left(\frac{\pi}{L_z} \right)}_{\text{volume / nodo}}} \Leftrightarrow$$

polarização

$$\Leftrightarrow n^\circ \text{ nodos} = \frac{k^2 dk}{\pi^2} \cdot L_x \cdot L_y \cdot L_z$$

$$\text{daqui: } \frac{n^\circ \text{ nodos}}{\text{volume}} = \frac{k^2 dk}{\pi^2} = \frac{8\pi \cdot v^2 \cdot dv}{c^3}$$

Na cavidade, a B.D.:

volume da cavidade

$$n^\circ \text{ nodos} = \frac{8\pi \cdot v^2 \cdot \Delta \nu_{\text{meio}}}{c^3} \cdot \underbrace{(0,15 \text{ m}) \cdot \pi \cdot 10^{-6} \text{ m}^2}_{\text{volume da cavidade}} =$$

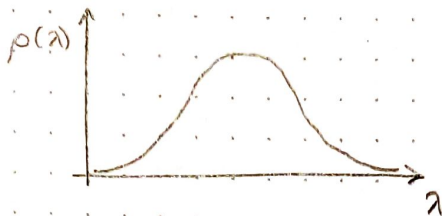
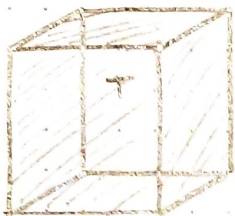
$$= \underline{\underline{148 \times 10^6 \text{ nodos}}}$$

acontece que:

$$\left| k = \frac{2\pi \cdot v}{c} \quad \text{e} \quad dk = \frac{2\pi}{c} \cdot dv \right|$$

___/___/___

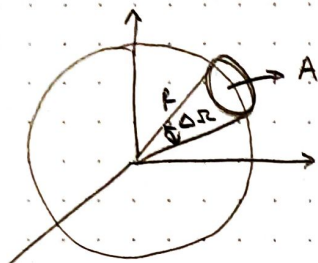
Lei de Planck → radiação do corpo



$\rho(\nu)$ → densidade espectral $\left(\frac{\text{energia}}{(\text{volume})(\text{frequência})} \right)$

$$\rho(\nu) d\nu = \frac{8\pi \cdot \nu^3}{c^3} \cdot d\nu \cdot \underbrace{h\nu}_{\text{energia / fóton}} \cdot \frac{1}{e^{h\nu/k_B T} - 1}$$

$$\left. \frac{1}{e^{h\nu/k_B T} - 1} \right\}$$

Ângulo Sólido3D

$$\Delta\Omega = \frac{A}{R^2}$$



2D

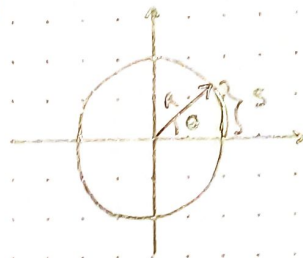
$$\theta = \frac{s}{R}$$

↓

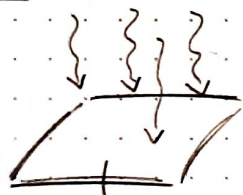
$$2\pi R = C$$

↓

Comprimento da circunferência



$$\frac{4\pi R^2}{R^2} = 4\pi \text{ (steradians)}$$

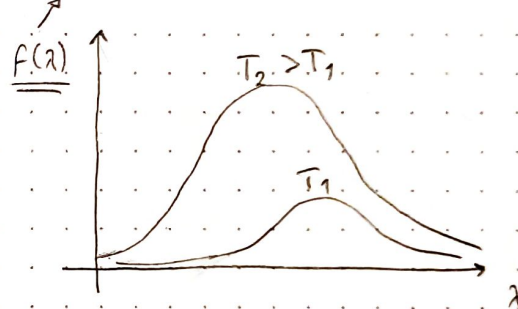
Problemas3.7.

"pedaço de asfalto"

$$I = 1000 \frac{\text{W}}{\text{m}^2}$$

$$\frac{c}{4} \cdot \int_0^\infty f(\nu) d\nu = \sigma \cdot T^4$$

espectro



Uma vez que $I = \sigma_{SB} \cdot T^4 \rightarrow 1000 = \sigma_{SB} \cdot T^4 \Rightarrow T = \left[\frac{1000}{5.67 \times 10^{-8}} \right]^{1/4}$

$\left[\frac{\text{W}}{\text{m}^2} \right] \quad \quad \quad \left[\sigma_{SB} \right] = \left[\frac{\text{W}}{\text{m}^2 \cdot \text{K}^4} \right]$

$T = 364 \text{ K } (91^\circ \text{C})$

3.9.

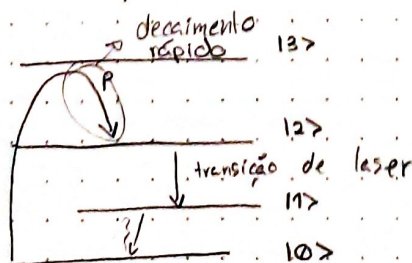
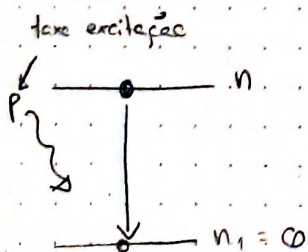
$$\frac{dN_2}{dt} = -\frac{1}{c} \cdot [B_{21} \cdot N_2 - B_{12} \cdot N_1] \cdot I_\nu \cdot S(\nu) - A_{21} \cdot N_2$$



$$\frac{dn}{dt} = -a \cdot n \cdot q - f \cdot n + p$$

$n = \frac{\text{N}^\circ \text{ átomos no estado excitado}}{\text{volume}}$

$q = \text{N}^\circ \text{ fótons}$



(...)

3.10.

SI N/m²

$$1 \text{ atm} = 1,01325 \times 10^5 \text{ N/m}^2$$

$$1 \text{ Torr} = 1 \text{ mm Hg}$$

$$760 \text{ Torr} = 1 \text{ atm}$$

gás ideal

nº de átomos

$$P.V = N \cdot k_B \cdot T$$

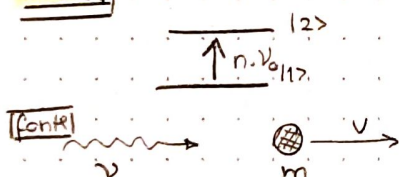
$$k_B = 1,38 \times 10^{-23} \text{ J/K}$$

$$\frac{N}{V} = \frac{P}{k_B \cdot T} \Leftrightarrow$$

$$\Leftrightarrow \frac{(1,01325 \times 10^5 \text{ N/m}^2) \cdot (\text{atm}/760) \cdot \text{Torr}}{1,38 \times 10^{-23} \text{ J/K} \cdot T [\text{K}]} = \frac{N}{V} \Leftrightarrow$$

$$\Leftrightarrow \frac{N}{V} = 9,65 \times 10^{24} \frac{P [\text{Torr}]}{T [\text{K}]} \leftarrow \underline{\underline{C \cdot q \cdot d.}}$$

3.12



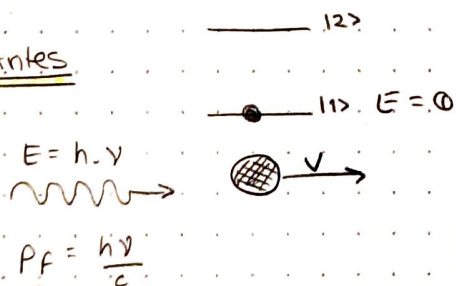
Considerando um fóton de frequência ν ,
energia $h \cdot \nu$ e momento linear $\frac{h \cdot \nu}{c} = p_{\text{fóton}}$.

$$E^2 = \underbrace{m^2 \cdot c^4}_{\text{energia cinética}} + p^2 \cdot c^2$$

$$\nu \ll c$$

$$\downarrow$$

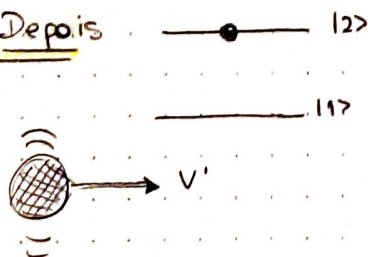
$$\frac{h\nu}{c} \ll m \cdot v$$

AntesConsidere a energia, E :

$$h\nu + \frac{1}{2} \cdot m \cdot v^2 = h\nu_0 + \frac{1}{2} \cdot m \cdot v_0^2$$

Considere o momento, $\vec{p}$:

$$\frac{h\nu}{c} + \frac{m \cdot v}{c} = m \cdot v' \rightarrow N' = \left(\frac{h\nu}{mc} + v \right)$$

Depois

Assim:

$$h \cdot \nu + \frac{1}{2} \cdot m \cdot \nu^2 = h \cdot \nu_0 + \frac{1}{2} \cdot m \cdot \left(\nu + \frac{h \cdot \nu}{mc} \right)^2$$

$$\frac{1}{2} \cdot m \cdot \nu^2 + \frac{1}{2} \cdot m \cdot 2 \cdot \nu \cdot \frac{h \cdot \nu}{mc} + \frac{1}{2} \cdot m \cdot \left(\frac{h \cdot \nu}{mc} \right)^2$$

$$\nu = \nu_0 + \nu \cdot \left(\frac{\nu}{c} \right) + \frac{1}{2} \cdot \frac{h \cdot \nu^2}{mc^2}$$

$$\nu \sim 3 \times 10^{12} \text{ m/s}$$

$$c \sim 3 \times 10^8 \text{ m/s}$$

$$h \nu = 2 \text{ eV}$$

$$630 \text{ nm}$$

$$e^- \cdot mc^2 \approx 0,5 \text{ MeV}$$

$$p \cdot mc^2 \approx 10^9 \text{ eV}$$



$$\nu \cdot \left[\frac{\nu}{c} + \frac{1}{2} \cdot \frac{h \cdot \nu}{mc^2} \right]$$

$$10^{-6}$$

$$10^{-9}$$

$$\nu \left[1 - \frac{\nu}{c} \right] = \nu_0$$

$$\nu = \frac{\nu_0}{1 - \frac{\nu}{c}} \approx \nu_0 \cdot \left[1 + \frac{\nu}{c} \right]$$

(. . .)

Problema 3.11

espaço onde tem CO_2 contido
 $\lambda = 10.6 \mu\text{m}$



$$\sigma(\text{CO}_2, \text{N}_2) = 1.20 \text{ nm}^2$$

seção de colisão

$$\sigma(\text{CO}_2, \text{O}_2) = 0.95 \text{ nm}^2$$

Nós queremos estimar o γ_c devido às duas colisões!

$$\gamma_c = \gamma_{\text{CO}_2-\text{N}_2} + \gamma_{\text{CO}_2-\text{O}_2}$$

$$\gamma_c = N \cdot \bar{v}_{\text{rel}} \cdot \sigma_{x-y}$$

$\downarrow$
 N: moléculas que você
 tem no CO_2
 $\downarrow$
 $\bar{v}_{\text{rel}}$: velocidade
 média relativa

$$\bar{v}_{\text{rel}} = \left[\frac{8RT}{\pi} \cdot \left(\frac{1}{M_x} + \frac{1}{M_y} \right) \right]^{1/2}$$

Pela Wikipedia:

$$\text{N}_2 \approx 78\% \approx 593 \text{ T}$$

$$\text{O}_2 \approx 21\% \approx 160 \text{ T}$$

$$1 \text{ atm} = 760 \text{ T}$$

$$N = 9.65 \times 10^{24} \cdot \frac{P(\text{Torr})}{T[\text{K}]} \cdot \frac{1}{\text{volume}}$$

$$N_{\text{N}_2} = 9.65 \times 10^{24} \cdot \frac{593}{300} \cdot \frac{1}{\text{m}^3}$$

$$\approx 1.9 \times 10^{25} \text{ m}^{-3}$$

$$N_{\text{O}_2} \approx 5.2 \times 10^{24} \text{ m}^{-3}$$

constante Boltzmann

$$N_A \cdot k_B = R = 8.31 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

$$T = 300 \text{ K}$$

$$\bar{v}_{\text{O}_2-\text{CO}_2} = 585 \text{ m/s}$$

$$\bar{v}_{\text{N}_2-\text{CO}_2} \approx 609 \text{ m/s}$$

$$\gamma_{\text{CO}_2-\text{O}_2}^{\text{col}} \approx 2.86 \times 10^9 \frac{\text{R}}{\text{s}} = N_{\text{O}_2} \cdot \bar{v}_{\text{O}_2}$$

$$\gamma_{\text{CO}_2-\text{N}_2}^{\text{col}} \approx 1.39 \times 10^{10} \frac{\text{R}}{\text{s}} = N_{\text{N}_2} \cdot \bar{v}_{\text{N}_2-\text{CO}_2} \cdot \sigma_{\text{N}_2-\text{CO}_2}$$

massas moleculares:

$$M_{\text{CO}_2} = 44 \text{ g/mol}$$

$$M_{\text{O}_2} = 32 \text{ g/mol}$$

$$M_{\text{N}_2} = 28 \text{ g/mol}$$

$$\gamma_T = \gamma_{\text{CO}_2-\text{O}_2} + \gamma_{\text{CO}_2-\text{N}_2} \approx 1.68 \times 10^{10} \text{ R/s}$$

$$\gamma_{\text{CO}_2}^{\text{col}} \approx 1.68 \times 10^{10} \text{ Rad/s}$$

$$\Delta \nu_{\text{colisões}}^{\text{Lorentziano}} = 2.76 \text{ Hz}$$

Lorentziano

$$N_{\text{N}_2} = 9.65 \times 10^{24} \cdot \frac{593}{300} \cdot \frac{1}{\text{m}^3}$$

largura Doppler

$$\Delta \nu_D = 2 \cdot \frac{V_0}{c} \sqrt{\frac{2k_B T \cdot \ln(2)}{m(\text{kg})}} =$$

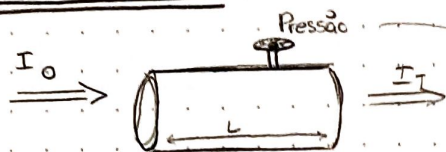
Comp. de onda pequeno $\rightarrow$ Desvio Doppler $\oplus$ pequeno

$$= 2.15 \times 10^5 \times \frac{1}{\lambda_0(\text{m})} \cdot \sqrt{\frac{T[\text{K}]}{M(\text{g/mol})}} \text{ MHz}$$

$$= 2.15 \times 10^5 \times \frac{1}{10600} \cdot \sqrt{\frac{300}{44}} \text{ MHz}$$

$$\Delta \nu_D \approx 53 \text{ MHz}$$

Problema 3.15



$$\frac{I_T}{I_0} = e^{-\alpha \cdot L}$$

$$\alpha(\nu) = \frac{\lambda^2}{8\pi} \cdot A_{21} \cdot \frac{g_2}{g_1} \cdot N \cdot S(\nu)$$

Quando a densidade é baixa, é proporcional ao volume dos átomos.

Limite Doppler Dominante:

$$S(\nu) = \frac{1}{\delta \nu_D} \cdot \left(\frac{4 \cdot \ln(2)}{\pi} \right)^{1/2} \cdot \exp \left[-\frac{4 \cdot (\nu - \nu_0)^2 \cdot \ln(2)}{\delta \nu_D^2} \right]$$

$$\delta \nu_D = \frac{2 \cdot \nu_0}{c} \cdot \sqrt{\frac{2 \cdot k_B \cdot T \cdot \ln(2)}{m}}$$

independente do N !

No limite em que as colisões são dominantes:

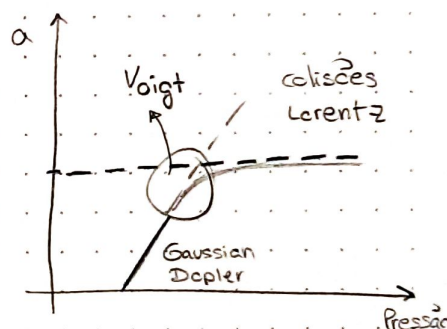
$$S(\nu) = \frac{\delta \nu / \pi}{(\nu - \nu_0)^2 + \delta \nu^2} \quad \text{Lorentz}$$

$$\delta \nu = \frac{A_{21}}{4\pi} + \frac{\gamma_c}{2\pi}$$

↑ tempo de vida

$$\gamma_c = N \cdot \bar{v}_{rel} \cdot \sigma$$

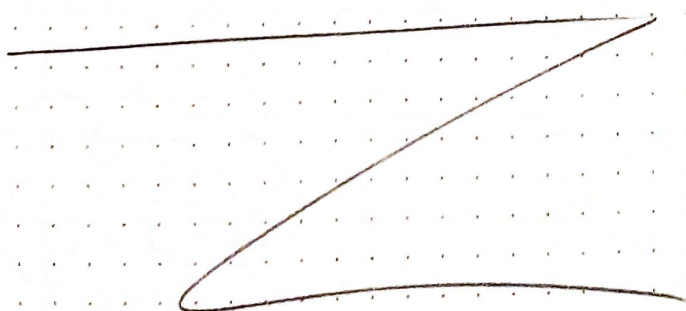
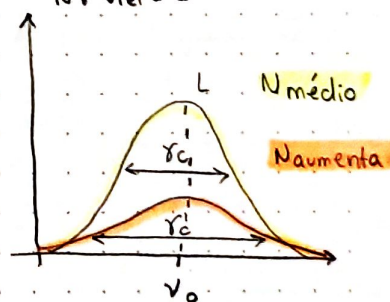
Se o N aumentar muito, o 1º termo será anulado



$$N \uparrow : \delta \nu \approx \frac{\gamma_c}{2\pi} \rightarrow S(\nu_0) = \frac{\delta \nu / \pi}{0 + \delta \nu^2} = \frac{\pi}{\delta \nu} = \frac{\pi}{N \cdot \bar{v}_{rel} \cdot \sigma}$$

$$a(\omega) = \frac{\lambda^2}{8\pi} \cdot A_{21} \cdot \frac{g_2}{g_1} \cdot N \cdot \frac{\pi}{N \cdot \sigma \cdot \bar{v}_{rel}}$$

↑ N elevado



mt. curto, na zona do UV !

Problema 3.19.

$$\text{He } 1S_0 \rightarrow 2P_1 \quad \lambda = 58,4 \text{ nm}$$

$$f = 0,28 \quad (\text{p.125})$$

(a) $A_{21} = ?$

$$A_{21} = \frac{g_1}{g_2} \cdot \frac{2\pi^2 \cdot e^2}{\epsilon_0 \cdot m \cdot c} \cdot f \cdot \frac{1}{\lambda_0^2} \Rightarrow \text{provém do modelo de Lorentz (+f) (eq. retirada do livro)}$$

$$= 1,8 \times 10^9$$

$$1S_0 \rightarrow n_j = 0 \quad g_1 = 1 \quad 2P_1 \quad n_j = -1 \quad n_j = 0 \quad n_j = 1$$

só existe uma possibilidade de conseguir um átomo a excitar

<https://physics.nist.gov/>

$$\lambda = 58,4339 \text{ nm}$$

$$A = 1,789 \times 10^9 \text{ R/s} \quad \text{radianos por segundo}$$

$$e = 1,6 \times 10^{-19} \text{ C} \quad c = 3 \times 10^8 \text{ m/s}$$

$$m_e = 9,1 \times 10^{-31} \text{ kg} \quad \epsilon_0 = 8,85 \times 10^{-12} \text{ N.m}^2/\text{C}^2$$

(b) $a(\nu_0) = \lambda_0^2 \cdot \frac{A_{21}}{8 \cdot \pi} \cdot \frac{g_2}{g_1} \cdot N \cdot S(\nu)$

↑
coeficiente de absorção
↓

$$\delta \nu_D = 2,15 \times 10^5 \cdot \frac{1}{\lambda_0 [\text{nm}]} \cdot \sqrt{\frac{T [\text{K}]}{\pi [\text{g/mol}]}} \text{ MHz}$$

Provém de

$$I(\omega) = I_0 \cdot e^{-a \cdot L}$$

baixa pressão (γ_0 pequeno)

Lorentz

$$T = 3000 \text{ K} \quad M_{\text{He}} = 4 \text{ g/mol}$$

$$\delta \nu_D^{\text{He}} = 31,9 \text{ GHz}$$

a 3000 K !

$$\delta \nu_D \approx \frac{A_{21}}{4\pi} + \frac{\gamma}{2\pi}$$

$$\approx \frac{1,8 \times 10^9}{4\pi} = 285 \text{ MHz}$$

Para obter o coeficiente de absorção no centro da linha:

Doppler

$$S(\nu) = \frac{1}{\delta \nu_D} \left(\frac{4 \cdot \ln(2)}{\pi} \right)^{1/2} \cdot \exp \left[\frac{-4 \cdot (\nu - \nu_0)^2 \ln(2)}{\delta \nu_D^2} \right]$$

$$n_j = -1 \quad n_j = 0 \quad n_j = 1 \quad g_2 = 3$$

$$n_j = 0 \quad g_1 = 1$$

$$S(\nu_0) = \frac{1}{\delta \nu_D} \cdot \left(\frac{4 \cdot \ln(2)}{\pi} \right)^{1/2}$$

$$a(\nu_0) = \lambda_0^2 \cdot \frac{A_{21}}{8\pi} \cdot \frac{g_2}{g_1} \cdot N \cdot \frac{1}{\delta \nu_D} \left(\frac{4 \cdot \ln(2)}{\pi} \right)^{1/2}$$

$$\lambda_0 = 58,4 \text{ nm} ; g_2 = 3, g_1 = 1 ; A_{21} = 1,8 \times 10^9 \text{ R/s}$$

___/___/___

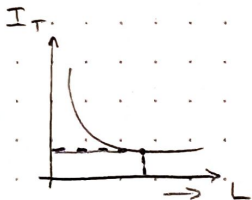
Para $P = 1 \text{ Torr}$ $N = 9,65 \times 10^{24} \frac{P (\text{Torr})}{T}$

a. $T = 300 \text{ K}$ $N \approx 3,2 \times 10^{22}$

$a(\nu_a) = 6,9 \times 10^{-15} \text{ m}^2$

comprimento absorção $(1/c) \frac{1}{a} \approx 1,45 \text{ p.m.}$

$\approx 25 \lambda_0$



28 / 10 / 2022

Problema 4.2

$$\frac{d}{dt} (n_2 + q_v) = - \left[\Gamma_2 + A_{21} \right] \cdot n_2 + p + \frac{c}{2L} (1 - \sigma_1 \cdot \sigma_2) \cdot q_v$$

Considerando o caso particular:

$$\underline{\underline{\Gamma_2 + A_{21}}} = \frac{c}{2L} (1 - \sigma_1 \cdot \sigma_2) = \Gamma_{total}$$

Neste caso, vamos ter

sol. equação homogênea

$$\frac{d}{dt} [N_2 + q_0] = - \Gamma_{total} \cdot (n_2 + q_0) + P$$

$$- \frac{d(N_2 + q_0)}{dt} = - \Gamma_{total} (n_2 + q_0)$$

sol. particular $(n_2 + q_0) = \frac{P}{\Gamma_{total}}$

$$\frac{d(n_2 + q_0)}{dt} - \Gamma_{total} \cdot (n_2 + q_0) = 0$$

Solução $n_2 + q_0 = A e^{-\Gamma_{total} t}$

(o o o)